

The Hong Kong University of Science and Technology

Optimizing Decisions for Personal and Business Development (IEDA 1250)

Summer, 2026

Instructor

- Prof. Bo YANG (yangb@ust.hk)

Class schedule

- Monday, Wednesday, Friday, 9:00 – 12:20

Tutorial

- Monday, Wednesday, Friday, 12:30 – 13:10

Teaching assistant

- Ms. Hanyan DUAN (hduanag@connect.ust.hk)
- Mr. Cho Yin NG (cyngba@connect.ust.hk)
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Course Description

Optimization plays a central role in enhancing decision making – be it in traditional engineering and business settings or in today’s more dynamic world powered by advanced AI techniques. This course introduces the core modeling and solution ideas behind optimization, with an emphasis on how to translate real problems into mathematical models and interpret solutions for actionable insights. The course covers linear programming, integer programming, probability, game theory, queueing theory, and dynamic programming, each of which offers a different perspective on how decisions can be modeled and analyzed. These topics are illustrated through applications in operations and analytics, including inventory management, production planning, service systems, and financial decision making. The course is intended to develop both practical modeling skills and a foundation for further study in optimization and decision analytics.

Intended Learning Outcomes

By the end of this course, students should be able to

- formulate introductory mathematical models for real-world decision problems, using tools such as linear programming, integer programming, game theory, queueing theory, and dynamic programming;
- interpret model outputs and communicate solution insights clearly in practical decision-making contexts;
- use quantitative reasoning to compare alternative decisions in personal, business, and engineering applications.

The course is introductory in nature and emphasizes modeling, interpretation, and applications rather than advanced mathematical proofs or technical derivations.

Recommended Textbooks

The textbooks are optional but recommended for students with particular interests in optimization. They contain broader and deeper contents than those covered in class. In particular, the third textbook contains rigorous mathematical proofs for many optimization algorithms (i.e., explanations on “why” the algorithms work).

- *Introduction to Operations Research*, by Hillier, Frederick S., and Gerald J. Lieberman, McGraw-Hill, 2015.
- *Operations and Supply Chain Management*, by F. R. Jacobs and R. Chase (13th edition), McGraw-Hill, 2018
- *Convex Optimization*, by Boyd, Stephen, 2004

This course will loosely follow the first textbook. There is no need for students to go through all materials to obtain decent scores. A list of tentative topics is included in the end of this syllabus.

Prerequisites

No prerequisite is required, as the course serves as an introduction to optimization.

Course Web Page

A web page will be available for this course on Canvas. The web page contains announcements, lecture notes, homework assignments and solutions. Slides will also be posted on Canvas before each lecture.

Grading

The final grade will be determined using the following weights:

1. Two Individual Assignments 20%
2. Midterm Exam 40%
3. Final Exam 40%
4. **Class Participation 0%**

Grading will be lenient in this course. In other words, requests for regrading will **NOT** be considered unless there are obvious grading errors. If students insist on regrading, all questions will be regraded for fairness.

Assignments

There will be two homework assignments. The first one will cover topics 1 - 4 (see tentative topics in the end of the syllabus); the second one will cover topics 5 - 7). Students will have a week to finish each of them. The specific time of releasing the homework depends on the progress.

Students are expected to work individually on these two assignments. A soft copy of the homework should be uploaded via Canvas on or before the due date. Late submissions will **NOT** be accepted, as the solution will be posted right after the due time.

If you use generative AI tools for learning or drafting, you must ensure the submitted work is your own. Using tools to generate complete solutions and submitting them as your own work is not allowed, and will result in zero grade for that particular assignment.

Exams

The midterm and final exams are 80 minutes exams. The midterm exam will be held during class after the topic “Game Theory”, depending on the progress. The final exam will be taken in the last class.

Class Participation

Please come to the class fully prepared. This will maximize the gain from the class. Regular attendance and participation in all classes will be helpful for border cases.

Policies

As a member of the HKUST community, students are expected to meet the highest standards of academic behavior. Please review the [university statement on academic integrity](#). On homework assignments, high-level collaboration, like discussion on methods to solve homework problems, is permitted. However, sharing solutions or numerical answers is not allowed and is considered cheating. Sharing solutions or cheating on exams will result

in a zero grade on that assignment or exam, and related university policies will be strictly enforced.

Tentative Topics

We will walk through the following topics. Coding skills are not required, though there are “programmings” in the topics.

1. Introduction
 - Motivating examples
2. Deterministic decision making: Introduction to linear programming
 - Motivating examples
 - Linear programming model
 - Graphical solution
 - Duality theory
 - Commercial solver: Excel and Gurobi
3. Deterministic decision making: Introduction to integer programming
 - Motivating examples
 - Integer programming model
 - Branch and bound approach
 - Modeling techniques
 - Commercial solver: Excel and Gurobi
4. Probability basics
 - Definitions such as probability, random variables, and cumulative distribution function
 - Discrete probability distributions
5. Making decisions with other players: Introduction to game theory
 - Motivating examples
 - Definitions
 - Dominating policies
 - Minimax theorem
 - LP approach
6. Decision making in service oriented processes: Introduction to queueing theory

- Motivating examples
- Basic metrics and parameters to a queueing system
- Little's law
- M/M/1 queue
- PASTA
- M/M/k queue

7. Sequential decision making: Introduction to dynamic programming

- Motivating examples
- Backward induction
- Shortest path problem
- Recourse allocation problem
- Production planning
- Brief introduction to Markov decision processes

8. Decision making in investment (optional)

- Brief introduction to risk neutral probability, replication, binomial tree, hedging, and valuation via real options approach